Need to know

What you need for each exercise. The T numbers point to the matching entries in the toolkit in `how-to-solve.pdf`. The full explanations are in `explanations.pdf`.

For all of them

- O means "grows no faster than" ($\leq$), Ω means "grows at least as fast as" ($\geq$), and Θ means both.
- Constant multipliers and small values of n do not count.
- Log rules: $\log(ab) = \log a + \log b$ and $\log(a^k) = k \log a$.

By exercise

0.1: Compare f and g *T1, T2, T3, T6, T7, T8, T9, T10*

- **Ratio test:** simplify f/g and see where it goes as n gets huge. If it goes to 0, then $f = O(g)$. If it goes to ∞, then $f = \Omega(g)$. If it settles at a positive constant, then $f = \Theta(g)$.
- **Growth ladder,** slowest to fastest: constants, $\log n$, $\sqrt{n}$, n, n^2, 2^n, $n!$.
- **Logs in different bases** differ only by a constant factor, so $\log 2n$ and $\log 3n$ are both $\Theta(\log n)$.
- **A different base in a power is not a constant factor:** 2^n and 3^n grow at different rates, because $(2/3)^n \to 0$.

0.2: Geometric series *T2, T3, T4, T8*

- **Closed form:** $1 + c + \cdots + c^n = \dfrac{c^{n+1} - 1}{c - 1}$. You get it by subtracting S from cS.
- **Which term matters:**
  - If $c < 1$, the first term dominates, so the sum is $\Theta(1)$.
  - If $c = 1$, the sum is $n + 1$ ones, so it is $\Theta(n)$.
  - If $c > 1$, the last term dominates, so the sum is $\Theta(c^n)$.

0.3: Fibonacci *T2, T5, T6, T11, T16*

- **Induction with two base cases,** because F_n depends on the two numbers before it.
- **Exponent split:** $2^{(n-1)/2} = 2^{n/2}/\sqrt{2}$.
- **The golden ratio:** $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$, and $\varphi^2 = \varphi + 1$. This is the step that makes the induction close.
- **The answer is** $c = \log_2 \varphi \approx 0.694$**.** Ω allows a constant in front of the bound, but not a bigger base.

1.1: Three digits in base b *T4, T5, T12*

- The largest digit in base b is $b-1$, and "100" in base b is b^2.
- **Check the worst case:** $3(b-1) < b^2$, which follows from $(b-1)(b-2) \geq 0$.

1.2: Binary vs decimal length *T4, T6, T12, T13*

- **Number of digits** of N in base B is $\lfloor \log_B N \rfloor + 1$.
- **Change of base:** $\log_2 N = \log_2 10 \cdot \log_{10} N$, and $\log_2 10 \approx 3.32$.
- **Handle** $d = 1$ **on its own.** The number 9 (1001_2) is the tight case.

1.3: Depth of a d-ary tree *T4, T6, T8, T13, T14*

- **What the letters mean:** n is the number of nodes, D is the depth, and d is the most children any node can have.
- **Level** k **holds at most** d^k **nodes.** Adding up the levels gives the 0.2 sum, so $n < d^{D+1}$, which means $D > \log_d n - 1$.
- **Exact minimum depth:** $D = \lceil \log_d(1 + n(d-1)) \rceil - 1$.

1.4: $\log(n!) = \Theta(n \log n)$ *T4, T6, T9*

- **Trap** $n!$ **between two powers:**
  - Upper bound: $n! \leq n^n$.
  - Lower bound: $n! \geq (n/2)^{n/2}$. Throw away the small half of the factors, and every factor left is at least $n/2$.
- **Take logs** to turn each power into a multiple of $n \log n$.

1.5: Harmonic sum $= \Theta(\log n)$ *T4, T8, T16*

- **Group the terms** into blocks between powers of 2. There are about $\log_2 n$ blocks.
- **Round the denominators down:** each block is at most 1.
- **Round the denominators up:** each block is at least $\frac{1}{2}$.

1.9 and 1.10: Modular arithmetic *T5, T15*

- **Translate first.** $x \equiv x' \pmod{N}$ means $x = x' + aN$ for some whole number a.
- **Goal in each proof:** show the difference is a whole number times N (or times M in 1.10).
- **1.9:** add the equations for the sum, and expand the brackets for the product.
- **1.10:** substitute $N = mM$. It only works in one direction: from N down to a divisor of N.

The two main methods

- **Trapping:** find a simple bound above and a simple bound below. Exercises 0.2, 1.2, 1.3, 1.4 and 1.5 work this way.
- **Translating into an equation, then doing algebra:** exercises 1.1, 1.9 and 1.10 work this way.

0.3 is the only one that needs induction.